Forecasting Tomorrow's Cognition or Profiling Today's Person? Trait-Level Affect and the Limits of Passive Sensing

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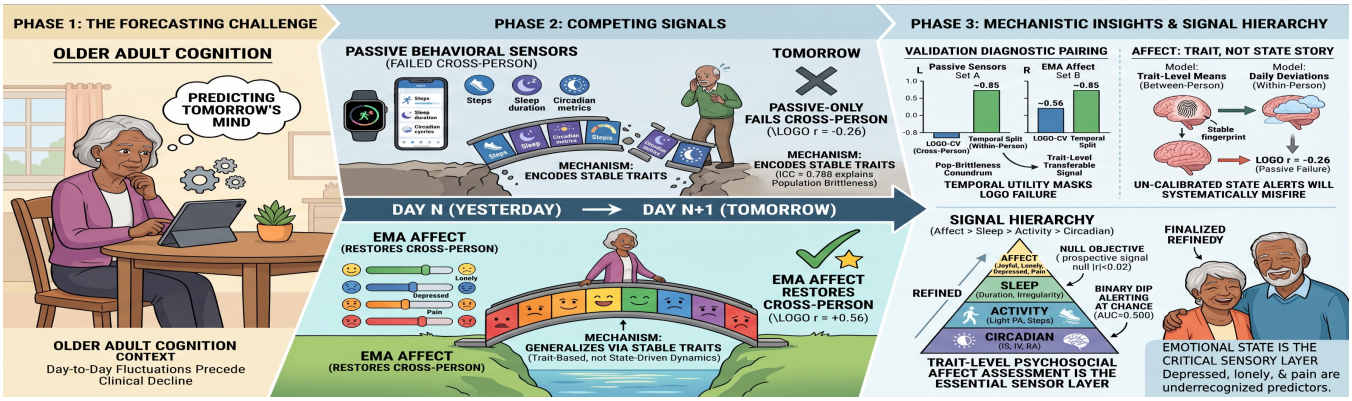


Figure 1: A visual overview of our next-day cognitive forecasting paradigm for older adults. The illustration contrasts the generalization failure of passive behavioral sensors with the indispensable predictive power of daily affect, culminating in our proposed signal hierarchy for ambulatory monitoring.

Abstract

Prospective next-day cognitive forecasting and cross-person generalizability remain understudied in older adults at dementia risk. In a 14-day EMA study (TecHealth Center, USC; N=102, ages 60–91), activPAL actigraphy, sleep logs, and a 15-item psychosocial battery yielded **1,376 Day N→N+1 pairs**; **84 participants** and **988 pairs** formed the ML sample. Passive-only features (sleep, activity, circadian) fail cross-person generalization (LOGO-CV; r=−0.26) while succeeding within-person (temporal r=0.85); ICC(1, 1)=0.785 reveals passive sensors encode stable between-person traits, not daily dynamics transferable to new individuals. FDR-corrected analysis identifies *depressed* (r=−0.459), *lonely* (r=−0.386), and *pain* (r=−0.368) as significant next-day predictors alongside standard mood items; objective micro-tests are uniformly null ($|r|$<0.02). Adding six EMA affect items substantially improves cross-person generalization (LOGO-CV r=0.561; MAE 9.95; permutation p<0.01). A within-vs-between-person decomposition shows this improvement is entirely trait-driven: within-person affect deviations yield r=−0.26 while between-person trait means yield r=+0.61. Critically, affect predicts *subjective* cognitive experience, clinically meaningful since subjective complaints precede dementia [7], not objective task performance ($|r|$<0.02). Binary dip detection yields AUC = 0.500, showing continuous forecasting success does not imply binary alert capability. We propose a four-tier Signal Hierarchy (Affect > Sleep > Activity > Circadian): trait-level affect is the strongest cross-person-generalizable modality, with direct implications for AI cognitive health monitoring.

CCS Concepts

• **Human-centered computing** → **Ubiquitous computing**; • **Applied computing** → **Health informatics**; • **Computing methodologies** → *Supervised learning by regression.*

Keywords

ecological momentary assessment; cognitive aging; passive sensing; next-day prediction; sleep behavior; affective state; personal informatics; older adults; wearables; machine learning

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1 Introduction

Cognitive decline in older adults advances along a trajectory punctuated by day-to-day fluctuations that impair independence and quality of life long before clinical diagnosis [7]. Early detection of these fluctuations, *before* they worsen, could enable preemptive action: adjusting caregiving intensity, reducing cognitive demands, or triggering clinical review. Traditional neuro-psychological assessment, conducted at months-long intervals, cannot capture this

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within-person variability; ecological momentary assessment (EMA) bridges the gap by delivering brief cognitive probes and self-reports multiple times daily in real-world environments [13]. Passive behavioral signals from wearable accelerometers, sleep diaries, and smartphone sensors are attractive complements to EMA because they require no active self-report. Prior work has linked such signals to depression and mood [3, 13], and Chikersal et al. [4] demonstrated prospective depression-onset prediction with passive phone data. Yet next-day *subjective cognitive* forecasting in older adults at dementia risk remains largely unaddressed. Prior studies predict the same-day (concurrent) cognitive state; whether Day N behavioral and affective patterns can reliably forecast Day N+1 performance is unknown. Critically, the gap is evaluative: no prior study has tested cross-person generalizability in this population alongside within-person forecasting. Preemptive action requires advance warning, not same-day detection; equally unexamined is *cross-person generalizability*, whether a model trained on some individuals transfers to new, unseen ones, a prerequisite for scalable deployment. We address both gaps with a prospective, lagged forecasting study using the **ECHO Lab Healthy Aging EMA dataset** [15]: 102 older adults, activPAL actigraphy, sleep logs, cognitive micro-tests, and a 15-item psychosocial EMA battery. Our analysis pursues three research questions:

RQ1. Do passive behavioral sensors (sleep, actigraphy, circadian) generalize to *unseen* individuals, and what mechanism explains any failure relative to within-person temporal forecasting?

RQ2. Which signal modalities most strongly predict next-day cognitive performance, and does the full 15-item psychosocial battery reveal predictors beyond standard mood items?

RQ3. Does baseline Subjective Cognitive Decline (SCD) status moderate behavioral-cognitive associations, and can Day-N signals detect impending low-cognition episodes?

The following are our Contributions:

(1) A **prospective lagged forecasting paradigm** with dual ecological validation (LOGO-CV and temporal split, permutation N=200, bootstrap N=1,000) directly testing cross-person generalizability in 102 older adults and 1,376 day-pairs.

(2) An **expanded bivariate predictor landscape**: in the 15-item EMA battery, *depressed*, *lonely*, and *pain* ($|r|>0.37$) rival standard valence items (bivariate only; Machine Learning (ML) Set B uses the 6-item companion battery); objective micro-tests are null ($|r|<0.02$), informing future ML feature selection.

(3) A **four-tier Signal Hierarchy** (Affect > Sleep > Activity > Circadian, in this cohort) grounded in ICC = 0.785 and a within-vs-between-person affect decomposition: passive sensors and within-person affect deviations both fail cross-person generalization (r=−0.26); affect generalizes primarily through stable between-person trait characteristics (trait-only model: r=+0.61), with direct implications for AI cognitive health monitoring design.

To our knowledge, this is the first study to demonstrate a complete reversal between temporal and cross-person validation in prospective subjective cognitive forecasting among older adults at dementia risk, the strongest single piece of evidence that passive monitoring pipelines require trait-level psychosocial calibration before cross-person deployment.

2 Related Work

2.1 EMA and Passive Sensing for Mental and Cognitive Health

EMA enables repeated, in-situ sampling of cognition and mood in natural environments [13]. The HCI community has linked passive behavioral signals to depression and mood: Wang et al. [13] demonstrated that smartphone data predicts depressive symptoms under participant-level cross-validation; Canzian and Musolesi [3] modeled depression trajectories from GPS traces; and Chikersal et al. [4] prospectively predicted the onset of depression from passive wearable data, the strongest methodological precedent for our lagged design. Cross-person evaluation schemes reveal systematic accuracy gaps masked by within-person temporal splits [4, 14], precisely the contrast we operationalize. Ambulatory cognitive micro-tests [6] capture within-person variability invisible to clinic-based assessment; yet previous work predominantly predicts same-day state or depression categories, leaving prospective next-day *subjective cognition* forecasting in older adults at dementia risk largely absent. To our knowledge, no prior work has explicitly contrasted LOGO-CV against temporal validation in this population. This specific pairing is crucial because it distinguishes generalizability between persons and clinical utility between individuals. Our study builds on two companion papers from the same ECHO Lab cohort: Yang et al. [15] examined concurrent ambulatory behavioral-cognitive associations without a prospective lagged design, and the companion LLM study [9] predicted same-day cognitive septiles using large language models.

2.2 Design Foundations, Sleep, and Aging

EMA-plus-passive-sensing designs for daily mental health monitoring [4] inform our extension to cognitive aging, and behavior change frameworks [5] motivate actionable monitoring targets. Sleep modulates next-day cognition [11]; prospective coupling of passively recorded sleep and circadian metrics with next-day cognitive outcomes in community-dwelling older adults at dementia risk remains an open gap we address directly.

3 Methods

3.1 Study and Participants

The study was conducted at the **TecHealth Center, University of South Carolina** [9, 15]. Community-dwelling older adults (ages 60–91; mean 68.3, SD 7.1; 65% female; 95% with at least some college education) without dementia diagnoses participated in a two-week measurement burst (data collection January 2021–October 2022; IRB: University of South Carolina #Pro00104601). Eligibility required ability to walk independently & at least 1 Alzheimer's disease related dementia risk factor (family history, subjective memory complaint, or BMI ≥25). Four times per day at semi-random intervals (~2.5 hr apart) during customized waking hours, participants completed EMA surveys on a dedicated smartphone app (M2C2) [15] delivering:

(a) **subjective cognition**: mindsharpness and concentration ratings averaged into cognitive composite (0–100);

(b) **15-item affective assessment**: joyful, calm, energetic, feel good, in-control, present, thoughts/emotions, physical wellbeing,

feel energetic, stressed, tired, anxious, depressed, lonely, and pain (all 0–100 visual analogue scales);

(c) **sleep quality**

(d) brief cognitive micro-tests and games (Symbol Search reaction time, Go/No-Go d', Dot Memory accuracy).

Participants wore a research-grade activPAL micro4 accelerometer (PAL Technologies, UK) on the right thigh continuously throughout the study, providing minute-level sedentary, light-activity (LPA), and moderate-to-vigorous physical activity (MVPA) classification [15]. Prior to the 14-day burst, all participants completed the Rabin et al. [10] SCD battery, comprising 22 items across memory, language, visuospatial, executive, and mood domains; total score > 6, the Youden-optimal cut point for incident amnestic MCI risk [10], defined Cognitive Decline.

The full EMA dataset yielded 1,376 lagged day-pairs (102 participants, 1,487 assessments; cognitive composite mean 75.4, SD 15.8); the merged ML sample (EMA+sleep+actigraphy) comprised 84 participants and 988 pairs; the MLM used 79 participants; the SCD sub-sample comprised 60 participants (Normal n=27; Decline n=33).

3.2 Feature Engineering

For each participant-day n, Day N predictors were extracted across four modalities:

Sleep (4 features): From the sleep diary: total sleep duration (hours) and bedtime hour. For both, we computed a within-person (WP) deviation from the individual's study-period mean, reflecting nightly variation around a personal baseline. Person-mean sleep duration and bedtime standard deviation (irregularity proxy) served as between-person (BP) features.

Activity (3 features): From minute-level actigraphy aggregated to the day: total step count, active minutes (activity counts > 0), & intradaily variability (IV) of minute-level counts, a fragmentation index where higher values indicate more fragmented activity.

Affective state (15 items): EMA scores for all 15 mood/state items (joyful, calm, energetic, feel good, in-control, present, thoughts or emotions, physical wellbeing, feel energetic, stressed, tired, anxious, *depressed*, *lonely*, *pain*), each the mean of up to four daily prompts. For the ML feature sets, we retained the 6 items from the companion study [9] (joyful, anxious, stressed, tired, calm, energetic); the full 15-item set was used in all bivariate correlation and SCD-stratified analyses.

Demographics and cognitive baseline (3 features): Age, sex, and BMI as BP predictors. Today's cognitive composite WP deviation (relative to personal mean) as a within-person autoregressive signal.

Continuous features were person-mean-centered to yield WP deviations, with person means as BP predictors; person-level IS, IV, and RA (NPCRA) served as circadian features. Affect BP features (study-period person means) are treated as stable trait baselines computed from each participant's full record, analogous to a clinical intake measurement; only the next-day cognitive composite target is withheld per LOGO-CV fold.

3.3 Feature Sets and ML Pipeline

We defined four cumulative feature configurations to isolate modality contributions: **Set A** (Passive only: sleep + activity + demographics); **Set B** (A + affect EMA); **Set C** (B + same-day cognitive test scores); **Set D** (Autoregressive ceiling: B + today's cognitive composite, a conceptual upper bound using a same-day cognitive measure not available at prediction time in real deployment).

Seven model families were trained: Ridge, ElasticNet, SVR (RBF kernel), Gradient Boosting, Extra Trees, XGBoost, and LightGBM; all with z-scored features within each fold and default hyperparameters (no tuning, by design): a finding robust across Ridge, tree ensembles, and kernel methods is measurement evidence independent of model-specific optimization [4, 14].

3.4 Cross-Validation Schemes

Leave-One-Person-Out (LOGO-CV): Each fold withholds one participant's full record as test (84 folds; `LeaveOneGroupOut` from scikit-learn), strictly testing generalization to *unseen* individuals.

Temporal split: Days were sorted chronologically per participant; earliest 60% trained, latest 40% tested (*same people*, different time points), testing within-person temporal generalization.

3.5 Ecological Validation Battery

ICC decomposition. $\mathrm{ICC}(1,1)$ [12] from a one-way ANOVA on cognitive composite decomposes total variance into stable between-person differences and within-person fluctuation.

Permutation test. Two hundred shuffles of Day N+1 cognitive composite labels (Ridge, Set B, LOGO-CV) produced empirical null distributions of r and MAE. The observed ElasticNet model was compared against these nulls.

Bootstrap CIs. N=1,000 resamples (no model refitting) yielded 95% CIs on r, MAE, and TA10 (tolerance accuracy ±10 points).

FDR correction. All bivariate correlations were Benjamini–Hochberg corrected [2]; $^{*}p<0.05$, $^{**}p<0.01$, $^{***}p<0.001$ denote FDR-adjusted thresholds throughout.

SHAP and CVI. SHAP [8] values (Extra Trees, Set C) ranked features by mean absolute contribution; a cognitive "dip day" (Day N+1 cognitive composite >1 within-person SD below mean) was used for logistic LOGO-CV AUC analysis.

4 Results

We first decompose cognitive composite variance to motivate the within-person centering strategy (§ 4.1), then survey the full bivariate predictor landscape across all 24 Day N features (§ 4.2), test cross-person ML generalization under dual validation with permutation and bootstrap confirmation (§ 4.3), add convergent mixed-model and SHAP evidence (§ 4.4), and close with SCD-stratified moderation and binary cognitive-dip detection (§ 4.5–6).

4.1 Cognitive Variance Decomposition

$\mathrm{ICC}(1,1)$=0.785: 78.5% of cognitive composite variance is stable between-person trait; only 21.5% is within-person daily fluctuation. This explains *a priori* why passive sensors fail LOGO-CV (they encode stable trait, not daily dynamics) while succeeding temporally, and motivates within-person centering of all continuous features. Stated concretely, knowing *who* someone is accounts for nearly four times as much cognitive variance as knowing how they are doing on any given day, a ratio that foreshadows the LOGO-vs-temporal divergence before a single model is trained.

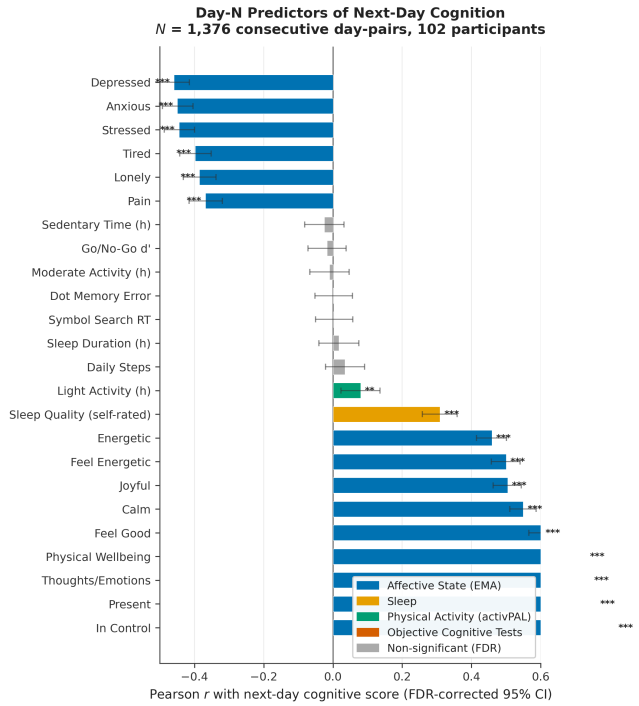


Figure 2: FDR-corrected Pearson r of 24 Day N features vs. next-day cognitive composite (N=1,376, 102 participants). Color encodes predictor category; gray = non-significant after FDR correction. $^{**}p<0.01$, $^{***}p<0.001$.

4.2 Bivariate Correlations: FDR-Corrected Day-N Predictors

Figure 2 shows FDR-corrected correlations for all 24 Day N predictors (N=1,376, 102 participants). Positive wellbeing items span *In Control* (r=+0.812) to *Joyful* (r=+0.504; all $p<0.001$). Negative affect is symmetrically broad: *Depressed* (r=−0.459), *Anxious* (r=−0.450), *Stressed* (r=−0.444), *Tired* (r=−0.398), *Lonely* (r=−0.386), and *Pain* (r=−0.368; all $p<0.001$) extend standard distress to somatic burden and psychosocial isolation. Sleep quality is the sole significant non-affective predictor (r=+0.309); all circadian metrics and objective cognitive micro-tests are null ($|r|<0.03$). Raw r's are dominated by stable between-person trait variance (ICC = 0.785); cross-person generalizability is assessed separately via LOGO-CV.

4.3 Cross-Person Generalization: Feature-Set Comparison and Ecological Validation

Table 1 reports the best-model results under LOGO-CV for each feature set, alongside the best temporal-split result. Under LOGO-CV, passive-only features (Set A) yield r=−0.264, worse than a person-mean baseline, confirming that sleep and activity signals alone cannot generalize to unseen individuals. Adding EMA affect (Set B, best: ElasticNet) substantially improves cross-person predictive power: LOGO-CV r=0.561, MAE = 9.95, R^2=0.310, TA10 = 58.0 % (bootstrap 95% CI: r [0.44, 0.67]). A WP/BP decomposition reveals the mechanism: WP-only affect yields LOGO-CV r=−0.26 (indistinguishable from passive failure); BP-trait-only yields r=+0.61, confirming Set B's benefit is entirely trait-driven (§ 6). Adding same-day

Table 1: Best-model performance per feature set under LOGO-CV and temporal split (Temp).

FS	CV	Model	r	MAE	R^2	TA10
A	LOGO	ENet	−0.264	13.54	−0.14	0.44
	Temp	LGBM	+0.845	6.37	n/a	0.80
B	LOGO	**ENet**	**+0.561**	**9.95**	**0.31**	**0.58**
	Temp	ET	+0.851	6.38	n/a	0.78
C	LOGO	ENet	+0.549	10.16	0.29	0.56
	Temp	ET	+0.853	6.48	n/a	0.78
D	LOGO	Ridge†	+0.908	4.66	0.83	0.90

A=passive; B=A+6 affect; C=B+cog. tests; D†=B+same-day cognitive composite. ENet=ElasticNet; ET=ExtraTrees; LGBM=LightGBM.

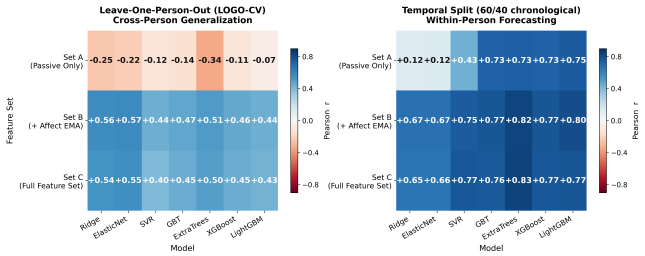


Figure 3: Pearson r across seven models and three feature sets under LOGO-CV (left) and temporal split (right). Red = negative r; blue = positive.

cognitive test scores (Set C) yields marginal further gain (r=0.549, ElasticNet).

Permutation test. Under 200 label shuffles (Ridge, Set B, LOGO-CV), the null r distribution had mean = −0.016 (SD = 0.044). The ElasticNet model (r=0.561) exceeded all 200 Ridge-null permutations (empirical $p<0.01$; null MAE mean = 12.54 vs. observed = 9.95); because ElasticNet outperforms Ridge on Set B, this comparison is conservative, confirming genuine cross-person signal.

LOGO-CV vs. temporal split. Figure 3 juxtaposes the two schemes: Set A diverges starkly (LOGO r=−0.26 vs. temporal r=+0.85), the central mechanistic finding (§ 5).

4.4 Convergent Evidence: MLM and SHAP

MLM (79 participants, REML) identifies three significant independent predictors: sleep duration WP deviation (β=−0.736, p=0.011), joyful (β=+0.121, $p<0.001$), and anxious (β=−0.094, p=0.004); short nights (<6 h) independently reduce next-day cognitive composite by 3.6 points (d=−0.231, p=0.034). SHAP values (Extra Trees, Set C) converge on the same story: *Joyful* leads (mean |SHAP|=0.216), followed by Tired (0.105), BMI (0.101), Age (0.091), Stressed (0.081), and Anxious (0.079); all passive actigraphy features fall at or below 0.035, corroborating Tier 3.

4.5 SCD-Stratified Analysis

Affect-cognition associations hold in both SCD groups but are attenuated in the Decline group: Joyful r=0.469 (Normal) vs. 0.403 (Decline); Depressed r=−0.482 vs. −0.284; light activity significant only in Normal (r=0.215, $p<0.001$). The attenuation is theoretically informative: elevated cognitive complaints may reduce the precision of subjective affect reports, weakening the trait-cognition coupling that Set B relies on, which suggests SCD status should

stratify model thresholds in deployed systems rather than a single uniform cut-point.

4.6 Cognitive Vulnerability Index (CVI)

A "dip day" (Day N+1 cognitive composite >1 SD below personal mean) occurred in 16.1% of pairs (159/988). No Day N feature distinguished dip from non-dip profiles after FDR correction (all $|d|<0.12$, $p>0.17$); logistic LOGO-CV yielded AUC = 0.500. This null is mechanistically predicted: the WP decomposition shows within-person deviations carry no cross-person signal (LOGO-CV $r=-0.26$), so day-level patterns cannot reliably flag personal dip days. Continuous forecasting success ($r=0.561$) does not imply binary alert capability; longer lag windows and HRV inputs are the natural next step.

5 Proposed Signal Hierarchy: A Four-Tier Empirical Hierarchy

Convergent evidence supports a four-tier Signal Hierarchy (Figure 3): in this cohort, *Tier 1 is the only modality to generalize across individuals*; Tiers 2–4 contribute within-person only, a key design consideration for population-level deployment.

Tier 1: Affective and Psychosocial State. Nine of 15 EMA items reach FDR significance bivariate ($p<0.001$); *depressed*, *lonely*, & *pain* add channels from psychosocial isolation and somatic burden absent from standard batteries, & are priority candidates for future ML sets. The 6-item ML battery (Set B) is the only modality to show positive cross-person generalization in this cohort (r [−0.26 to +0.561], via stable between-person traits); MLM & SHAP converge on joyful ($\beta=+0.121$, SHAP 0.216) & anxious ($\beta=-0.094$) as dominant channels.

Tier 2: Sleep Behavior. WP sleep duration ($\beta=-0.736$, $p=0.011$) and bedtime irregularity ($r=-0.075$, $p=0.019$) [11] contribute independently; a short night (<6 h) reduces next-day cognitive composite by 3.6 points ($d=-0.231$).

Tier 3: Activity Patterns. Light activity is the sole significant predictor ($r=+0.080$, $p=0.008$); sedentary time is null for next-day mean cognition yet predicts within-person cognitive variability in the same cohort [15].

Tier 4: Circadian Rhythms. All NPCRA metrics (IS, IV, RA) are uncorrelated with next-day cognitive composite ($|r|<0.03$, $p>0.72$), likely reflecting intact circadian organization in this non-demented community sample [11].

6 Discussion

6.1 Affect as the Leading Modality

Our decomposition experiment settles a fundamental mechanistic question: affect's LOGO-CV benefit derives entirely from *stable between-person trait characteristics*, not daily mood fluctuations. A LOGO-CV model restricted to within-person affect deviations yields $r=-0.26$ (indistinguishable from passive-only failure), while a model using only between-person trait means yields $r=+0.61$. What Set B encodes is "chronically joyful individuals have higher cognitive baselines," a stable profile property, not "above-average-joyful days predict better next-day cognition." The extended 15-item battery deepens this picture: *depressed*, *lonely*, and *pain* ($|r|>0.37$) capture additional trait-level burden channels, namely psychosocial isolation and somatic attentional depletion, that are absent from standard monitoring batteries yet strongly linked to accelerated cognitive aging [7].

The deployment implication reframes this work as *baseline cognitive risk profiling*: affect supports identifying whose trait-level baseline warrants closer monitoring, not generating day-to-day dip alerts. Systems firing on "low joyful today" without person-specific calibration will misfire, since within-person deviations carry no cross-person signal. Architecturally, this means affect measurement should function as a *population-level intake screener*, administered at onboarding to establish an individual's trait profile rather than a real-time state monitor issuing daily alerts based on momentary mood changes.

Scope: subjective cognition, not objective performance. Because both the cognitive composite and affect predictors are EMA self-reports, bivariate correlations partly reflect self-report coherence; however, LOGO-CV and WP/BP decomposition show that the cross-person signal reflects stable trait affect rather than momentary covariation. We do not claim affect predicts objective cognitive performance: self-reported cognition showed meaningful prediction (LOGO-CV r = 0.561), whereas objective micro-test scores did not ($|r|<0.02$ across tasks). This scope boundary is clinically meaningful because subjective cognitive complaints often precede dementia-related decline [7].

6.2 Validation Scheme, Brittleness, and AI Risk

The Set A divergence ($r=-0.26$ LOGO-CV vs. $r=+0.85$ temporal) illustrates a systematic population-brittleness risk: temporal accuracy conceals cross-person failure on new users [1, 14], precisely the evaluation gap this workshop foregrounds. We recommend the LOGO-CV-vs-temporal pairing as a standard diagnostic in ambulatory cognitive health research. Three concrete AI deployment risks follow from our findings.

(a) Inherited cross-person failure. An LLM health-coach trained only on passive behavioral traces inherits the LOGO-CV failure ($r=-0.26$) for genuinely new users; temporal accuracy ($r=+0.85$) belongs only to individuals already in training. Any AI cognitive monitoring pipeline must collect psychosocial affect from new users before cross-person inference is warranted.

(b) Trait-to-state hallucination. Because Set B's benefit is entirely trait-driven (BP-only: $r=+0.61$; WP-only: $r=-0.26$), an AI system may infer "good cognitive health" from a habitually positive affect profile, conflating stable personality with real-time cognitive state and generating alerts that reflect *who someone is* rather than *how they are today*.

(c) Fairness risk. The cohort is 72% White and 95% college-educated, and any model trained here encodes trait profiles specific to this demographic. Deployment on more diverse or under-resourced communities risks systematic disparities in alert equity without fairness auditing [7].

6.3 Limitations and Future Work

These findings should be interpreted as proof-of-concept. The study used a single-site, demographically homogeneous sample (N = 84; 72% White; 95% college-educated), a two-week measurement window, and modest SCD subgroup sizes, which limited generalizability

and stratified analysis. Future work should pursue multi-site replication, longer observation windows, additional physiological signals such as HRV, and adaptive micro-EMA designs to mitigate EMA nonresponse and passive-only generalization failure.

7 Conclusion

Passive sensing encodes *who* an older adult is, capturing stable between-person traits that account for 78.5% of subjective cognitive variance (ICC = 0.785), but fails to predict how a new, unseen individual will perform tomorrow. Contrasting LOGO-CV against temporal validation is the diagnostic pairing that exposes this gap and should be standard practice in ambulatory cognitive health research. Six EMA affect items address the cross-person gap that passive sensing cannot close (r improves from -0.26 to 0.561). A within-vs-between-person decomposition shows this improvement operates through stable trait-level affect (trait-only: r=+0.61; WP-only: r=−0.26), informing how AI cognitive health systems should use EMA data. The expanded 15-item battery identifies *depressed*, *lonely*, and *pain* as underrecognized trait-level predictors; objective micro-tests carry no prospective cross-person signal. The four-tier Signal Hierarchy translates to a concrete directive: integrate a brief psychosocial probe as the critical first tier for cross-person cognitive risk stratification. Binary dip detection (AUC = 0.500) remains the open frontier as continuous forecasting success does not imply binary alert capability.

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